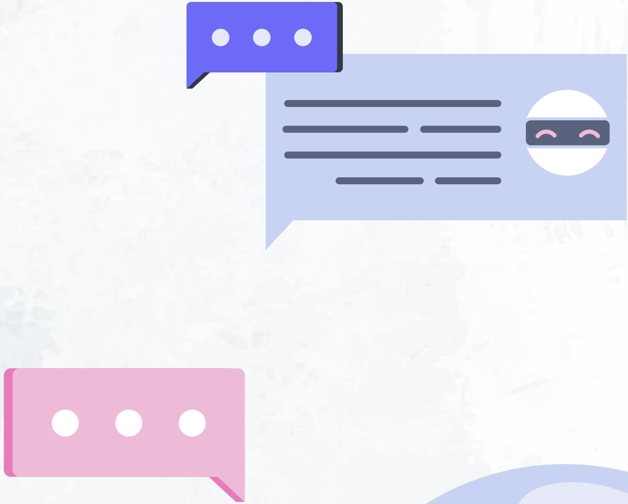
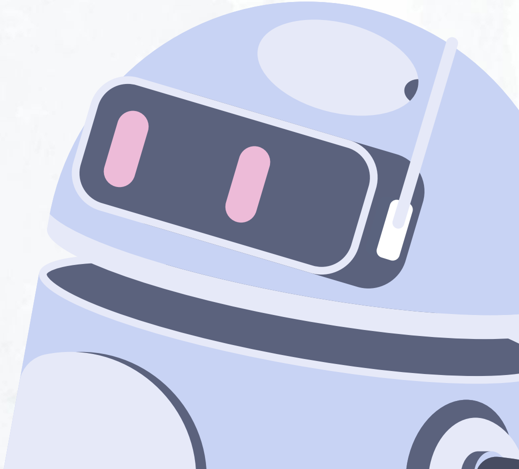
LLM Sentiment project





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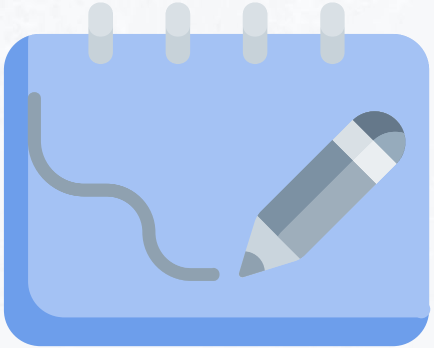
Introduction & Motivation

Our goal is to automatically analyze opinions in user reviews. Manual analysis is slow and overwhelming as data grows. Aspect-Based Sentiment Analysis (ABSA) helps us find what people talk about (e.g., food, service, location) and whether their opinion is positive, negative, or neutral.

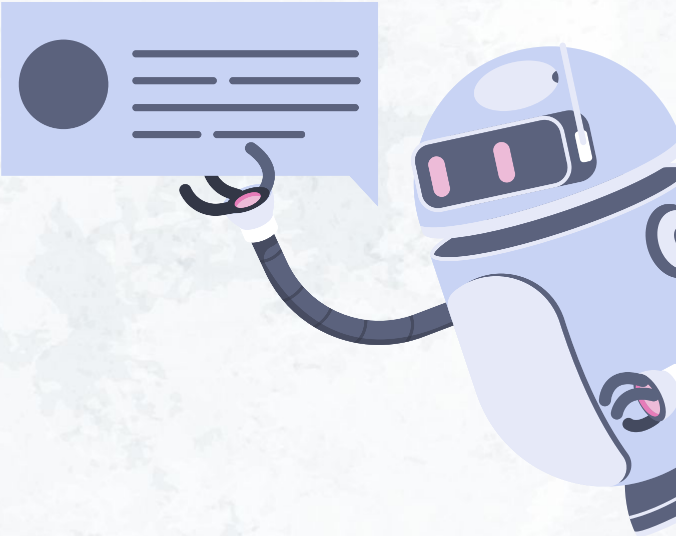


Project Objectives

Implement and compare three ABSA methods:

- 📌 Lexicon-based (fast, simple rules)
- 📌 Transformer-based (context-aware model)
- 📌 LLM-based (advanced, deep understanding)

Determine which is most accurate and efficient.

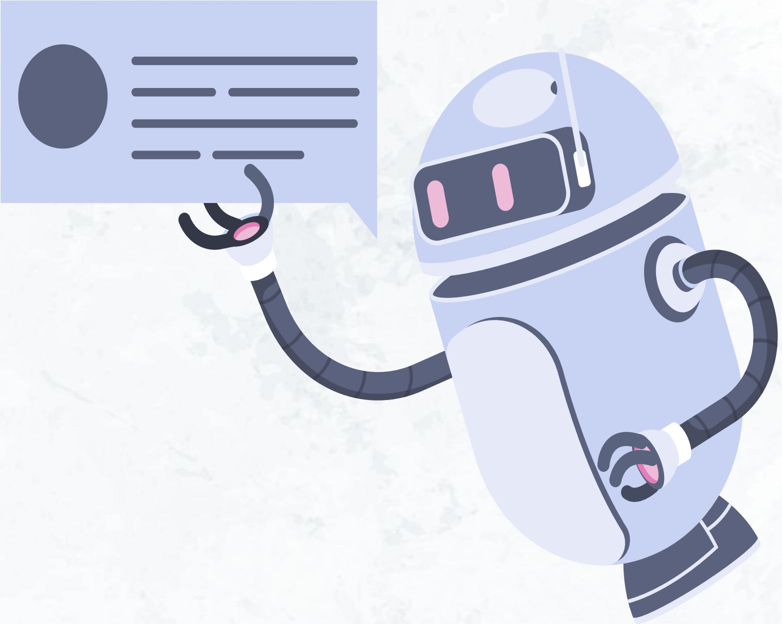


Methods Overview: LexiconABSA

- Rule-based using **SpaCy + VADER**.
- Extracts aspects using syntactic parsing.
- Finds sentiment words nearby (adjectives, verbs).

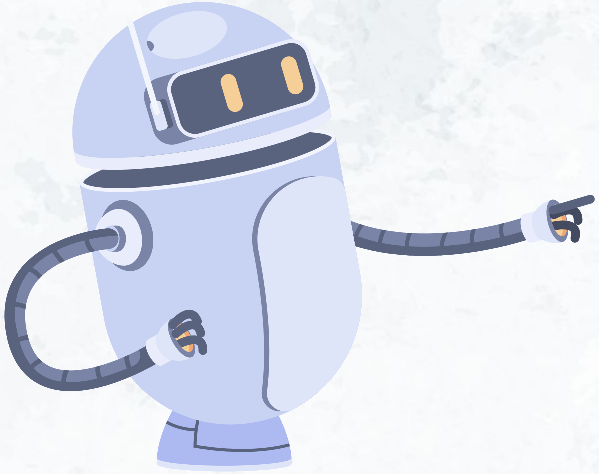
Methods Overview: TransformerABSA

- Aspect extraction with **gauneg/roberta-base-absa-ate-sentiment**
- Sentiment classification with **yangheng/deberta-v3-base-absa-v1.1**



Methods Overview: LLMABSA

- Uses a **Large Language Model** to do both tasks directly.
- Reads the whole text and infers aspects + sentiments together.



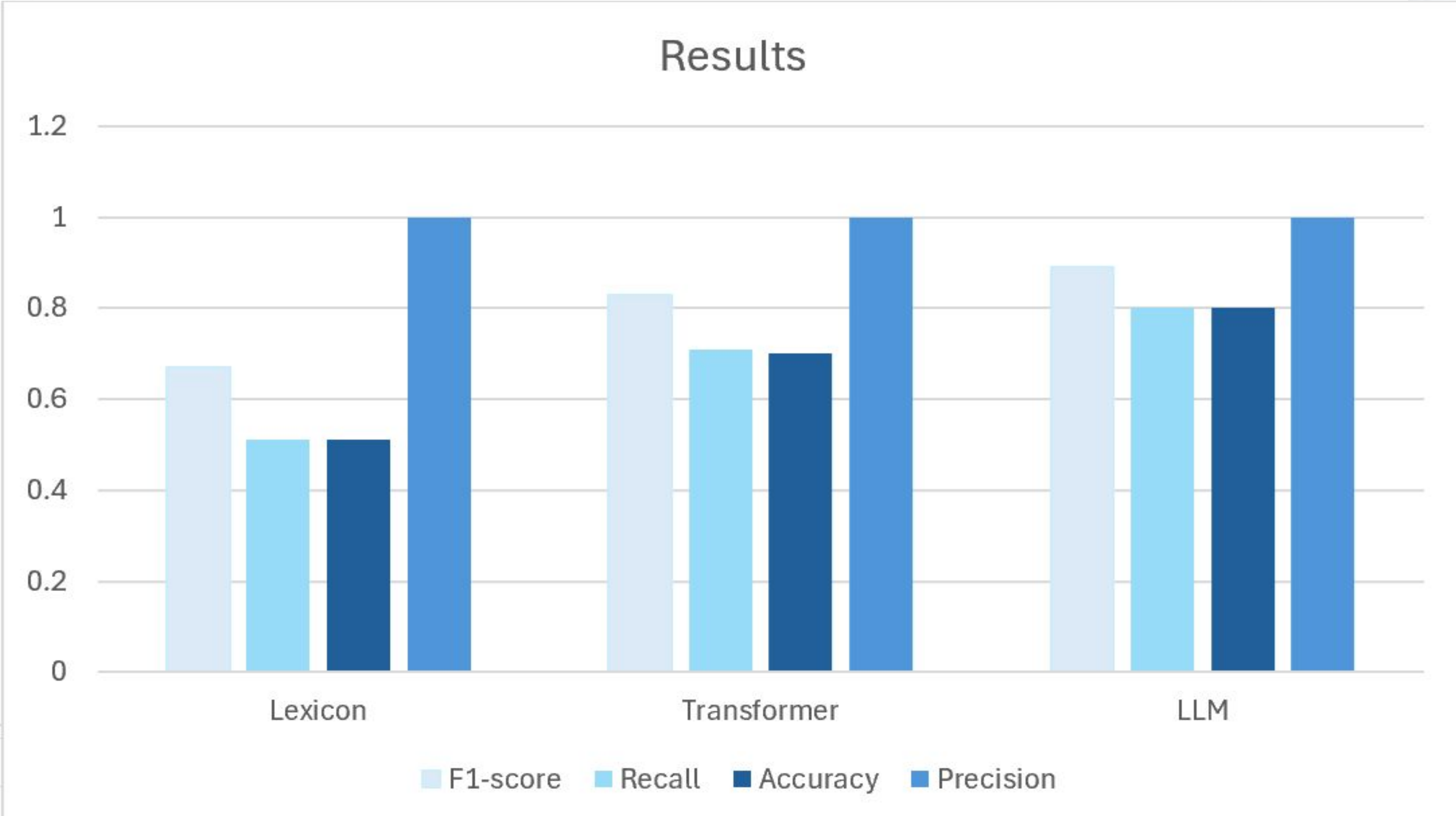
Example Review Analysis

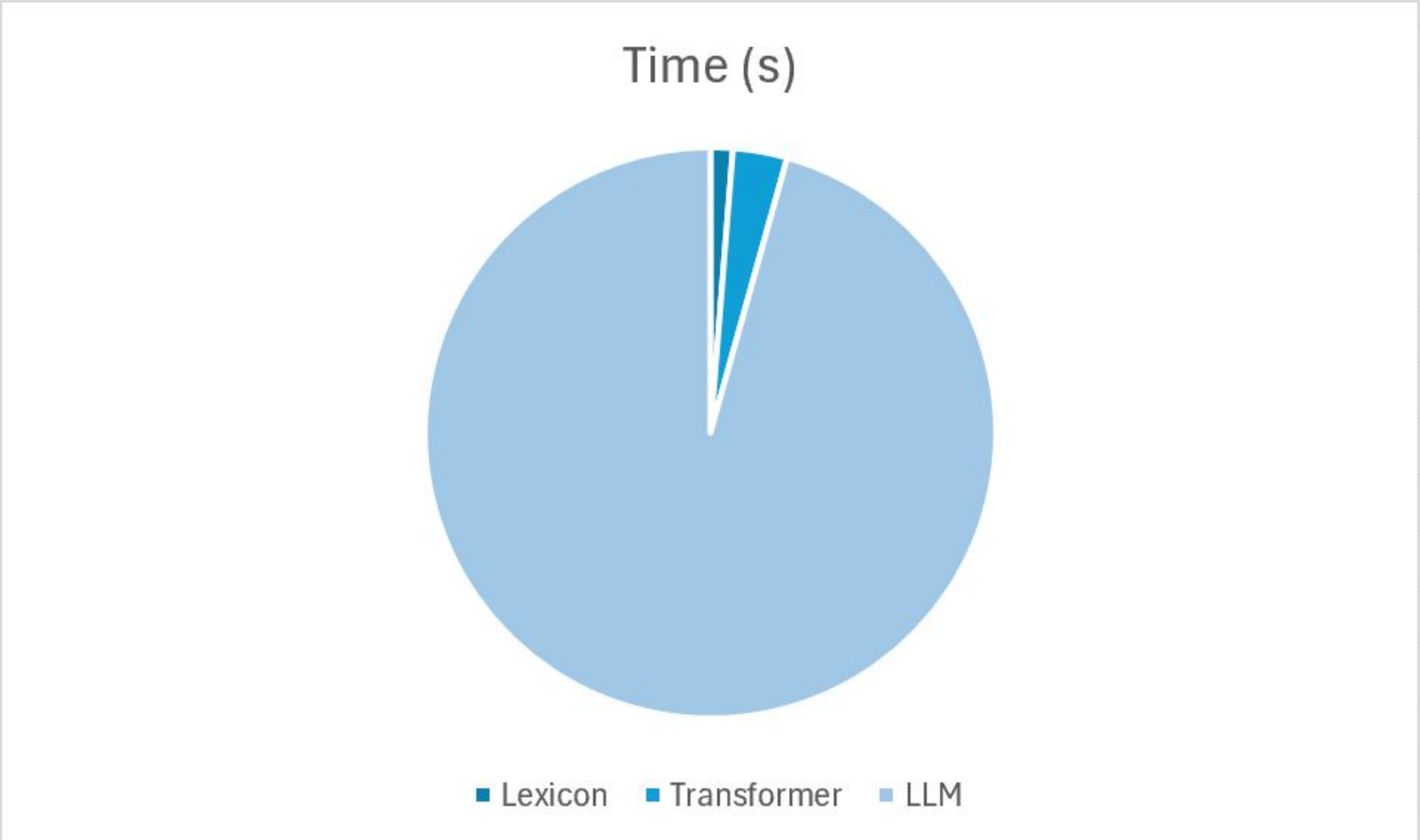
"The food was amazing but the service was slow."

- Lexicon: Finds 'food' (positive), misses weak sentiment for service .
- Transformer: Finds both aspects, captures more context.
- LLM: Understands even subtle or complex expressions.

Results

Model	Precision	Recall	F1-score	Accuracy	Avg Confidence	Avg Inference Time (s)
Lexicon	1.00	0.510	0.675	0.510	NaN	0.144
Transformer	1.00	0.706	0.828	0.706	NaN	0.342
LLM	1.00	0.804	0.891	0.804	0.327	10.644


Results
1.2
1
0.8
0.6
0.4
0.2
0
Lexicon
Transformer
LLM
F1-score
Recall
Accuracy
Precision


Time (s)
Lexicon
Transformer
LLM

Strength & Weakneses

- Lexicon: Fast, simple, but misses complex or indirect sentiment.
- Transformer: Balanced speed and accuracy.
- LLM: Most accurate, slowest, resource-intensive.

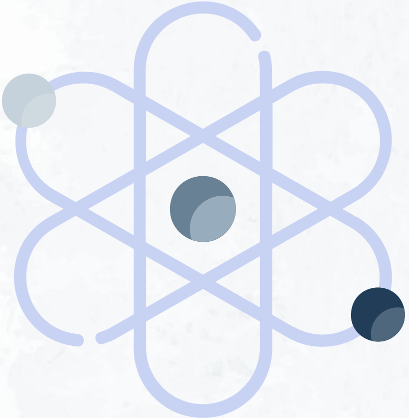
Use Case Recommendations

Lexicon for quick, large-scale trends

Transformer for business insights and balanced tasks.

LLM for high-accuracy research or subtle opinion mini

Key Challenges & Solutions

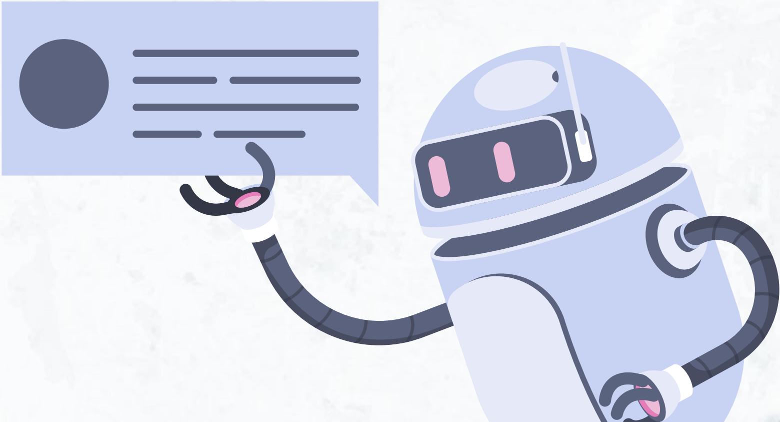


Sarcasm, implicit sentiment, and multi-word aspects are hard to detect.

Improved preprocessing, specialized models, and error handling helped address challenges.

Conclusion

Each model has strengths: Fast (Lexicon), Accurate (LLM), or Balanced (Transformer). Future work: Combine methods for even better results.



Q&A

Thank you for listening!

Any questions?